

# Valuation of Options - part 2

## Quantitative Finance

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# Assumptions Black-Scholes market (recap)

In class we will almost exclusively work with the 'Black-Scholes market'.

## Assumptions on price processes:

Asset price:

$$dS_t = \mu S_t dt + \sigma S_t dW_t, \quad S_0 = s_0, \quad \text{var}(W_1) = 1$$

Money Market Account:

$$dB_t = rB_t dt, \quad B_0 = 1$$

## Assumptions on market:

- frictionless trading
  - no transaction costs
  - trading in continuous-time possible
  - no restrictions on short sales and fractional positions
- borrowing rate = lending rate

- we are interested in price  $C_t$  of (European) option with payoff  $C_T = h(S_T)$
- we will discuss three methods to determine fair price (using continuous-time model for financial market):
  - Black-Scholes Partial Differential Equation
  - risk-neutral pricing
  - pricing kernel

**These slides discuss the Black-Scholes Partial Differential Equation approach.**

## Section 1

# The Black-Scholes Partial Differential Equation

# The problem

## Setup:

- Black-Scholes market (1 risky asset)
- we restrict to Markovian trading strategies:

$$\phi_t = \tilde{\phi}(B_t, S_t) = \phi(t, S_t) \text{ and } \psi_t = \tilde{\psi}(B_t, S_t) = \psi(t, S_t)$$

So we can write, for some function  $F$ ,

$$V_t = \phi(t, S_t)S_t + \psi(t, S_t)B_t = F(t, S_t)$$

## Question:

Which  $F$ 's correspond to **self-financing** trading strategies?

## Why interesting?

- consider option with payoff  $h(S_T)$  at maturity
- if we can find self-financing portfolio with  $F(T, S_T) = h(S_T)$  then no-arbitrage price of option at time  $0 \leq t < T$  is given by

$$C_t = F(t, S_t)$$

# Black-Scholes Partial Differential Equation

## Black-Scholes Partial Differential Equation:

$F$  corresponds to a self-financing trading strategy if and only if  $F$  is a solution to the Partial Differential Equation (equation with function  $G$  as variable)

$$\frac{\partial G}{\partial t}(t, s) + rs \frac{\partial G}{\partial s}(t, s) + \frac{1}{2} \sigma^2 s^2 \frac{\partial^2 G}{\partial s^2}(t, s) - rG(t, s) = 0 \quad \forall s > 0, t \in [0, T),$$

and in that case the positions in the self-financing trading strategy are given by:

$$\phi(t, s) = \frac{\partial F}{\partial s}(t, s),$$

and

$$\psi(t, s) = \frac{F(t, s) - \phi(t, s)s}{\exp(rt)}.$$

## Remarks:

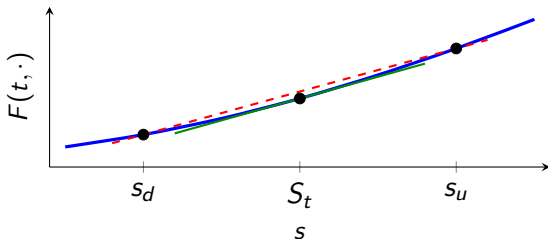
- $\mu$  does not play a role!
- using different models (i.e. SDEs) for  $B$  and  $S$  leads, in general, to different PDE!

# From the binomial tree to the delta

In the one-period binomial model (part 1) the replicating stock position was a *difference quotient*,

$$\phi = \frac{c_u - c_d}{s_u - s_d} = \frac{F(t + \Delta t, s_u) - F(t + \Delta t, s_d)}{s_u - s_d},$$

the slope of the value across the two stock outcomes. In the PDE it is  $\phi = F_s = \partial F / \partial s$ , the **delta**: as the step shrinks, the secant slope  $\rightarrow$  the tangent slope.



**secant** slope =  $\frac{c_u - c_d}{s_u - s_d}$  (binomial)  $\rightarrow$  **tangent** slope =  $F_s = \Delta$  (PDE).

# Derivation of the PDE

Which  $F$  are possible (when using self-financing strategies  $(\phi(t, S_t), \psi(t, S_t))$ )? We have

(1)  $V_t = F(t, S_t)$

(2)  $dV_t = \phi(t, S_t) dS_t + \psi(t, S_t) dB_t$

- apply Itô to (1) and insert  $dS_t \implies dV_t = a_t dt + b_t dW_t$
- insert  $dS_t$  and  $dB_t$  in RHS of (2)  $\implies dV_t = c_t dt + d_t dW_t$
- obtain system of equations

$$\begin{cases} a_t = c_t \\ b_t = d_t \end{cases}$$

- solving  $\implies$  conditions on  $F$  (PDE)



## Warning: sloppy notation

We will often use the following notations/abbreviations when working with PDEs:

- $F_S = (\partial F / \partial s) = (\partial F / \partial s)(t, S_t)$
- $F_t = (\partial F / \partial t) = (\partial F / \partial t)(t, S_t)$
- $F = F(t, S_t)$
- etc.

The notation  $F_t$  is somewhat confusing as it seems to refer to a stochastic process. And you argue that it would be a natural abbreviation for  $F(t, S_t)$ .

If you want to be clear and safe it could be better to just write down the partial derivatives fully.

# Derivation

We have

(1)  $V_t = F(t, S_t)$

(2)  $dV_t = \phi(t, S_t) dS_t + \psi(t, S_t) dB_t$

- apply Itô to (1) and insert  $dS_t = \mu S_t dt + \sigma S_t dW_t \implies$

$$\begin{aligned} dV_t &= F_S dS_t + F_t dt + \frac{1}{2} F_{SS} d[S, S]_t \\ &= \left( F_S \mu S_t + F_t + \frac{1}{2} F_{SS} \sigma^2 S_t^2 \right) dt + F_S \sigma S_t dW_t \end{aligned}$$

- insert  $dS_t$  and  $dB_t = rB_t dt$  in RHS of (2)  $\implies$

$$dV_t = (\phi_t \mu S_t + r \psi_t B_t) dt + \phi_t \sigma S_t dW_t$$

- hence  $\phi_t = \phi(t, S_t) = F_S(t, S_t) = F_S$
- and

$$F_S \mu S_t + F_t + \frac{1}{2} \sigma^2 F_{SS} S_t^2 = \phi_t \mu S_t + r \psi_t B_t$$

We need to prove

$$\frac{\partial F}{\partial t}(t, s) + rs \frac{\partial F}{\partial s}(t, s) + \frac{1}{2} \sigma^2 s^2 \frac{\partial^2 F}{\partial s^2}(t, s) - rF(t, s) = 0 \quad \forall s > 0, t \in [0, T)$$

On the previous slide we obtained

$$F_t + \frac{1}{2} \sigma^2 F_{SS} S_t^2 = r\psi_t B_t.$$

As  $\psi_t B_t = F - \phi_t S_t = F - F_S S_t$  we obtain the result.

# Alternative Derivation

- sell (write) and hold 1 option with price process  $C_t = F(t, S_t)$  (assumption!)
- let  $\phi_t$  be number of stocks at time  $t$  and  $\psi_t$  the position in the MMA
- yields portfolio value  $V_t = -C_t + \phi_t S_t + \psi_t B_t$
- this portfolio is self-financing if

$$dV_t = -dC_t + \phi_t dS_t + \psi_t dB_t$$

- using Itô:

$$dV_t = -F_S dS_t - F_t dt - \frac{1}{2} F_{SS} d[S, S]_t + \phi_t dS_t + \psi_t r B_t dt$$

- can only eliminate local risk ( $dW_t$ ) for  $\phi_t = F_S(t, S_t)$ !
- yields  $dV_t = \dots dt$  which has (locally) no risk, so rate of return is same as on  $B \implies dV_t = rV_t dt$
- hence

$$-F_t - \frac{1}{2} F_{SS} \sigma^2 S_t^2 + r\psi_t B_t = r(-F + F_S S_t + \psi_t B_t)$$

which yields Black-Scholes PDE

# Application

Given is a European option with payoff  $h(S_T)$  at expiration date/maturity  $T$ . How can we use the PDE to obtain the price of this option?

- Solve the PDE,

$$\frac{\partial G}{\partial t}(t, s) + rs \frac{\partial G}{\partial s}(t, s) + \frac{1}{2} \sigma^2 s^2 \frac{\partial^2 G}{\partial s^2}(t, s) - rG(t, s) = 0,$$

for all  $s > 0$ ,  $t \in [0, T)$ , under the boundary condition

$$G(T, s) = h(s) \text{ for all } s > 0.$$

- If  $F$  is solution to PDE satisfying the boundary condition, then no-arbitrage implies that the price of the option at time  $t \in [0, T)$  is given by  $F(t, S_t)$ .
- If you are lucky: 'closed-form' solution can be found
  - there is relation to well studied heat equation
- unlucky: use numerical techniques; see notebook

## Examples: closed-form solutions

- $h(S_T) = S_T$

$$F(t, s) = s$$

- European call option:  $h(S_T) = \max\{S_T - K, 0\}$

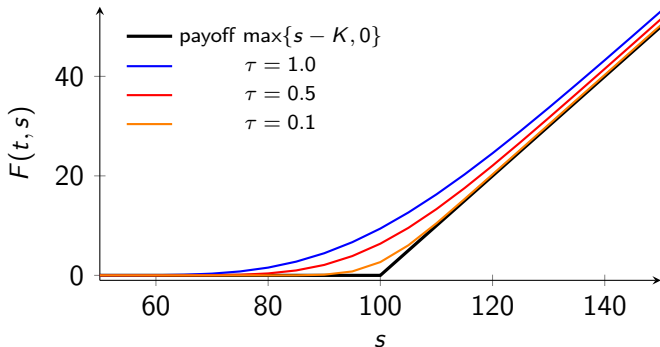
$$F(t, s) = s\Phi(d_1) - \exp(-r(T - t))K\Phi(d_2),$$

with  $d_1 = d_2 + \sigma\sqrt{T - t}$  and

$$d_2 = \frac{\log(s/K) + (r - 0.5\sigma^2)(T - t)}{\sigma\sqrt{T - t}}$$

# The call price as a function of $s$

The PDE solution  $F(t, s)$  for a European call ( $K = 100$ ,  $r = 3\%$ ,  $\sigma = 20\%$ ), for several times-to-maturity  $\tau = T - t$ :



As  $\tau \rightarrow 0$  the price collapses onto the payoff; the gap above it is the option's *time value*.

## Worked example: verifying a candidate solution

To check that a function  $F$  solves the B-S PDE, just substitute it. Two quick cases (recall  $F_t + rsF_s + \frac{1}{2}\sigma^2 s^2 F_{ss} - rF = 0$ ):

**(a) the stock,  $F(t, s) = s$ :** then  $F_t = 0$ ,  $F_s = 1$ ,  $F_{ss} = 0$ , so

$$0 + rs \cdot 1 + 0 - r s = 0. \quad \checkmark$$

**(b) a zero-coupon bond paying  $K$  at  $T$ ,  $F(t, s) = K e^{-r(T-t)}$ :** then  $F_s = F_{ss} = 0$  and  $F_t = rK e^{-r(T-t)} = rF$ , so

$$rF + 0 + 0 - rF = 0. \quad \checkmark$$

By *linearity* of the PDE, any combination  $as + bK e^{-r(T-t)}$  also solves it — consistent with such a stock+bond portfolio being self-financing.

(In the exercises you verify that the digital-call solution  $F(t, s) = e^{-r(T-t)} \Phi(d_2)$  solves the PDE.)



## Quiz

A portfolio has value process  $V_t = F(t, S_t)$  with

$$F(t, s) = 2s + 3K e^{-r(T-t)}.$$

- 1 Does  $F$  solve the Black-Scholes PDE?
- 2 Which self-financing strategy  $(\phi, \psi)$  does it correspond to?
- 3 Which claim does the portfolio pay at maturity  $T$ ?

## Examples: European digital call option

- payoff:  $h(S_T) = 1\{S_T \geq K\}$
- the B-S PDE can be solved explicitly:

$$F(t, s) = \exp(-r(T - t))\Phi(d_2),$$

with

$$d_2 = \frac{\log(s/K) + (r - 0.5\sigma^2)(T - t)}{\sigma\sqrt{T - t}}$$

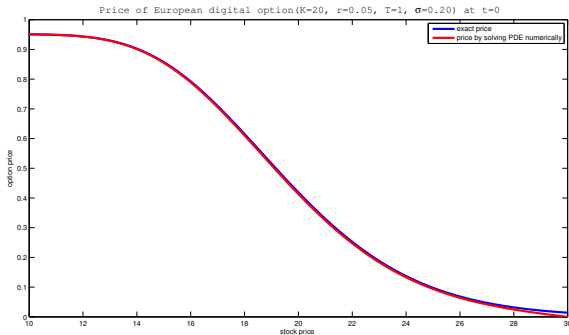
- Probability that option ends in-the-money:

$$\mathbb{P}\{S_T \geq K\} = \Phi\left(\frac{\log(s/K) + (\mu - 0.5\sigma^2)(T - t)}{\sigma\sqrt{T - t}}\right)$$

**In the exercise set you will be asked to verify that  $F$  is a solution to the B-S PDE.**

# Examples: European digital put option

Price of the option (at  $t = 0$ )  $F(0, s)$  - numerical solution compared to exact solution:

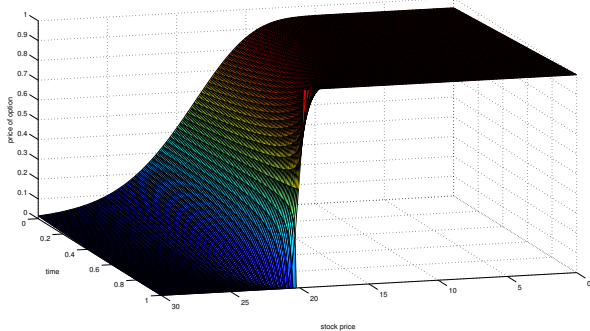


The Group Assignment might contain an exercise on solving the PDE numerically.

# Examples: European digital put option

Value function  $F(t, s)$ :

Price of option as function of time and stock price European digital option,  $K=20$ ,  $T=1$ ,  $r=0.05$ ,  $\sigma=0.20$  (exact solution)



# Solving the PDE numerically: finite differences

When there is no closed form, solve the PDE on a grid  $t_0 < \dots < t_N = T$  (step  $\Delta t$ ) and  $s_0 < \dots < s_M$  (step  $\Delta s$ ).  
Replace the derivatives by differences:

$$F_t \approx \frac{F(t + \Delta t, s) - F(t, s)}{\Delta t}, \quad F_s \approx \frac{F(t, s + \Delta s) - F(t, s - \Delta s)}{2 \Delta s},$$
$$F_{ss} \approx \frac{F(t, s + \Delta s) - 2F(t, s) + F(t, s - \Delta s)}{\Delta s^2}.$$

- start from the known terminal values  $F(T, s) = h(s)$  and step *backward* in time;
- the explicit scheme writes each  $F(t, s_i)$  as a (discounted) weighted average of its three neighbours  $F(t + \Delta t, s_{i-1})$ ,  $F(t + \Delta t, s_i)$ ,  $F(t + \Delta t, s_{i+1})$  — i.e. a *trinomial tree*.

An implementation — including the explicit/trinomial scheme and its stability condition — is provided in the accompanying notebook.

The explicit finite-difference scheme *is* a trinomial tree; the binomial tree of part 1 is its two-neighbour cousin. For the scheme to make sense the three weights must be **nonnegative** — they play the role of risk-neutral probabilities.

- nonnegative weights require  $\Delta t$  small relative to  $\Delta s^2$  (a *stability / CFL* condition); otherwise the scheme oscillates and the “probabilities” turn negative;
- the same constraint shows up in the tree: the log-price step must scale like  $\sigma\sqrt{\Delta t}$  (so  $u = e^{\sigma\sqrt{\Delta t}}$ ) to keep  $q \in (0, 1)$ .

**One picture:** numerical stability  $\Leftrightarrow$  nonnegative weights  $\Leftrightarrow$  a valid risk-neutral measure  $\Leftrightarrow$  no arbitrage. The PDE, the trees, and the grid are three views of the same object.

# From the Black-Scholes PDE to the heat equation

A change of variables makes the connection precise. Write  $\tau = T - t$  (time-to-maturity) and  $x = \log s$ , and set  $F(t, s) = v(\tau, x)$ . Using  $s \partial_s = \partial_x$  and  $s^2 \partial_{ss} = \partial_{xx} - \partial_x$ , the PDE turns into a *constant-coefficient* equation with an *initial* condition  $v(0, x) = h(e^x)$ :

$$v_\tau = \frac{1}{2} \sigma^2 v_{xx} + \left(r - \frac{1}{2} \sigma^2\right) v_x - r v.$$

Rescaling time as  $\tilde{\tau} = \frac{1}{2} \sigma^2 \tau$  and substituting  $v = e^{ax+b\tilde{\tau}} u$  with

$$a = -\frac{1}{2}(k-1), \quad b = -\frac{1}{4}(k+1)^2, \quad k = \frac{2r}{\sigma^2},$$

removes the first- and zeroth-order terms, leaving the pure **heat equation**

$$u_{\tilde{\tau}} = u_{xx}.$$

So pricing a European option is, after a change of variables, an initial-value problem for the heat equation.

## Food for thought

The B-S PDE comes with a *terminal* condition  $F(T, s) = h(s)$ , so it runs *backward* in time. With  $\tau = T - t$ ,  $x = \log s$  (and a discount/scaling factor) it becomes the *heat equation*

$$\frac{\partial u}{\partial \tau} = \frac{\partial^2 u}{\partial x^2}.$$

- The heat equation instantly *smooths* its initial data. What does this say about the smoothness of the price  $F(t, \cdot)$  for  $t < T$ , even when the payoff  $h$  has a kink (call) or a jump (digital)?
- Why is “backward in time, from maturity” the natural direction for valuation?



# Feynman-Kac Theorem

Let  $\alpha(t, x)$ ,  $\beta(t, x)$ , and  $k(t, x)$  be functions satisfying 'regularity conditions'. Consider the PDE

$$\frac{\partial G}{\partial t}(t, x) + \alpha(t, x) \frac{\partial G}{\partial x}(t, x) + \frac{1}{2} \beta^2(t, x) \frac{\partial^2 G}{\partial x^2}(t, x) - k(t, x) G(t, x) = 0,$$

for all  $x$  and  $t \in [0, T)$  with boundary condition  $G(T, x) = h(x)$  for all  $x$ .

Then the solution is given by:

$$G(t, x) = \mathbb{E} \left[ \exp \left( - \int_t^T k(s, X_s) ds \right) h(X_T) \mid X_t = x \right],$$

where  $X$  is a stochastic process, on the time interval  $[t, T]$ , defined via the SDE

$$dX_u = \alpha(u, X_u) du + \beta(u, X_u) dW_u$$

and with starting value  $X_t = x$ , and where  $W$  is a standard Brownian motion.

# Sketch of proof (Feynman-Kac)

Let  $F$  solve the PDE with  $F(T, x) = h(x)$ , and let  $X$  follow  $dX_u = \alpha(u, X_u) du + \beta(u, X_u) dW_u$  on  $[t, T]$  with  $X_t = x$ . Define the discounted process

$$M_u = \exp\left(-\int_t^u k(v, X_v) dv\right) F(u, X_u), \quad u \in [t, T].$$

With  $D_u = \exp(-\int_t^u k dv)$ , Itô's formula gives

$$\begin{aligned} dM_u &= D_u \left[ -k F du + (F_t + \alpha F_x + \frac{1}{2} \beta^2 F_{xx}) du + \beta F_x dW_u \right] \\ &= D_u \underbrace{\left( F_t + \alpha F_x + \frac{1}{2} \beta^2 F_{xx} - kF \right)}_{=0 \text{ (the PDE)}} du + D_u \beta F_x dW_u. \end{aligned}$$

So  $M$  has *no drift*: it is a (local) martingale, hence — under integrability conditions — a martingale. As  $M_t = F(t, x)$  and  $M_T = \exp(-\int_t^T k dv) h(X_T)$ ,

$$F(t, x) = \mathbb{E}[M_T \mid X_t = x] = \mathbb{E}\left[\exp\left(-\int_t^T k(v, X_v) dv\right) h(X_T) \mid X_t = x\right].$$

# Black-Scholes PDE reconsidered

The no-arbitrage price, at time  $t \in [0, T)$ , of a European option with payoff  $h(S_T)$  at expiration date  $T > 0$  is given by  $F(t, S_t)$  where  $F$  satisfies

$$\frac{\partial F}{\partial t}(t, s) + rs \frac{\partial F}{\partial s}(t, s) + \frac{1}{2} \sigma^2 s^2 \frac{\partial^2 F}{\partial s^2}(t, s) - rF(t, s) = 0 \quad \forall s > 0, t \in [0, T)$$

Feynman-Kac tells us that we have

$$F(t, s) = \mathbb{E} \left[ e^{-r(T-t)} h(S_T) \mid S_t = s \right],$$

where  $S$  is defined via the SDE

$$dS_u = rS_u du + \sigma S_u dW_u, \quad S_t = s,$$

where  $W$  is a standard Brownian motion.

# Uniqueness and the maximum principle

Two structural facts make “the” price well-defined and well-behaved.

- **Uniqueness.** Under mild growth conditions the B-S PDE together with the terminal condition  $F(T, s) = h(s)$  has a *unique* solution. Hence  $F(t, S_t)$  is unambiguous — the analytic counterpart of market *completeness*.
- **Maximum principle.** For this (parabolic) PDE a solution attains its extreme values on the boundary of the domain — here the terminal time and the  $s$ -boundaries. Two consequences:
  - *monotonicity*: if  $h_1 \leq h_2$  then  $F_1 \leq F_2$  (prices respect the ordering of payoffs);
  - *no-arbitrage bounds*, e.g.  $0 \leq F_{\text{call}}(t, s) \leq s$ .

Both are also immediate from Feynman-Kac:

$F(t, s) = \mathbb{E}[e^{-r(T-t)} h(S_T) \mid S_t = s]$  is linear and monotone in the payoff  $h$ .